

21 Survival Analysis

Suppose that we have collected a five-year medical study, in which patients have been treated for cancer. We would like to fit a model to predict patient survival time, using features (predictors) such as treatment, underlying conditions, etc. This is an example in which the outcome is **time until an event** occurs. Other examples are:

- Time from diagnosis to death
- Time from surgery to relapse
- Time to infection after vaccination
- Time to hospital discharge

A common feature that makes survival analysis different from ordinary regression is **censoring**. In our five-year study, some patients will still be alive at the end of follow-up. For these patients, the survival time is **right-censored**: we do not observe their event time, but we know they survived at least 5 years.

For each individual i , we suppose that there is a true **survival time** T_i or a true **censoring time** C_i . We observe either T or C , that is, we observe

$$Y_i = \min\{C_i, T_i\}$$

We also observe a status indicator

$$\delta = \begin{cases} 1 & \text{if } T \leq C \\ 0 & \text{if } T > C \end{cases}$$

- Assume we observe n pairs $(y_1, \delta_1), \dots, (y_n, \delta_n)$. - We further assume that the event time T is independent of the censoring time C . An example when this is violated occurs when males who are very sick are more likely to drop out of the study than females who are sick. In this case, comparison of survival time between male and female may wrongly suggest males survive longer.

The **survival function** is defined as

$$S(t) = P(T > t)$$

It tells us the probability of surviving beyond time t .

Example: BrainCancer dataset

The BrainCancer dataset (package ISLR2) consists of survival times of 88 patients diagnosed with brain cancer. It contains information about 8 variables: sex, diagnosis, location factor, karnofsky index, gross tumor value, stereotactic method, status and survival time in months.

- Let $d_1 < d_2 < \dots < d_K$ denote the K unique death times among the non-censored patients, and let q_k denote the number of patients who died at

time d_k .

$$= P(A, B) + P(A, B^c)$$

- Let r_k denote the number of patients alive and in the study just before d_k , these are the at **risk** patients.

By the law of total probability

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

$$S(d_k) = P(T > d_k) = P(T > d_k | T > d_{k-1})P(T > d_{k-1}) + \underbrace{P(T > d_k | T \leq d_{k-1})P(T \leq d_{k-1})}_{=0}$$

which can be simplified as:

$$S(d_{k-1})$$

$$S(d_k) = P(T > d_k | T > d_{k-1})P(T > d_{k-1}) = P(T > d_k | T > d_{k-1})S(d_{k-1})$$

This implies that

$$S(d_k) = P(T > d_k | T > d_{k-1}) \times \dots \times P(T > d_2 | T > d_1)P(T > d_1)$$

It is then natural to estimate

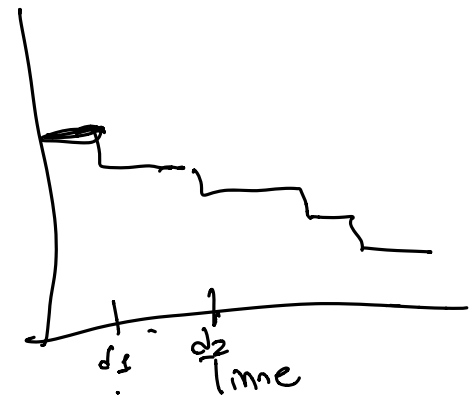
$$\hat{P}(T > d_j | T > d_{j-1}) = \frac{r_j - q_j}{r_j}$$

Kaplan-Meier

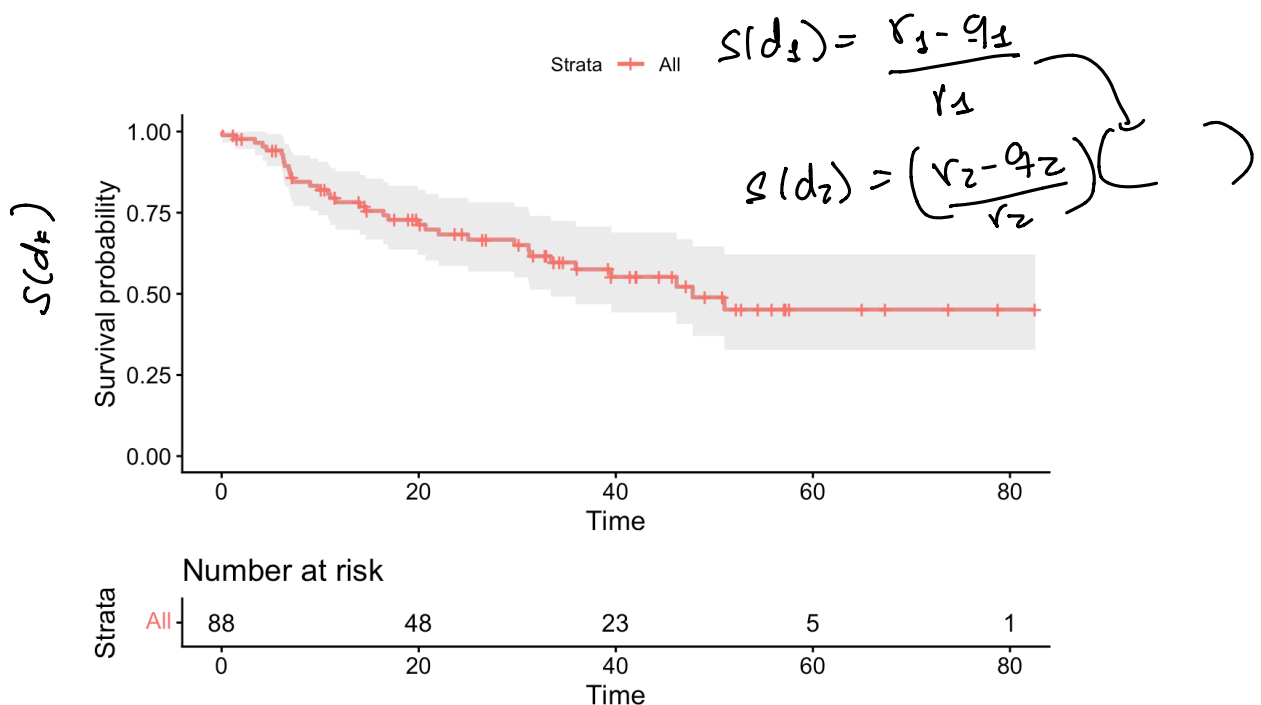
to obtain the ~~Kaplan~~-Meier estimator of the survival curve:

$d_j = j$ -th death-time

$$\hat{S}(d_k) = \prod_{j=1}^k \left(\frac{r_j - q_j}{r_j} \right)$$



```
library(ISLR2)
library(survival)
library(survminer)
fit.surv<-survfit(Surv(time,status)~1,data=BrainCancer)
ggsurvplot(fit.surv, data = BrainCancer, conf.int = TRUE, risk.l
```



```
##Alternatively
```

```
##plot(fit.surv,xlab="Months",ylab="Estimated Survival probability")
```

21.1 Logrank test

The logrank test is a nonparametric test for comparing whether two groups have the same survival function over time (equivalently, the same hazard function at all times under its standard formulation). It is widely used with right-censored time-to-event data.

Let $d_1 < d_2 < \dots < d_K$ denote the unique death times, and let $q_{1,k}$ and $q_{2,k}$ denote the number of patients in groups 1 and 2 who died at time d_k . Let $r_{1,k}$ and $r_{2,k}$ denote the number of patients in groups 1 and 2 who are in risk.

In order to test $E(X) = 0$, we construct a test statistic of the form:

$$W = \frac{X - EX}{\sqrt{\text{Var}(X)}}$$

Specifically,

$$W = \frac{\sum_{k=1}^K (q_{1,k} - \frac{r_{1,k}q_k}{r_k})}{\sqrt{\frac{q_k(r_{1,k}/r_k)(1-r_{1,k}/r_k)(r_k-q_k)}{r_k-1}}}$$

E[deaths in group 1]
 $q_k \times \left[\frac{r_{1,k}}{r_k} \right]$

When the sample size is large, this statistic has approximately a standard normal distribution that can be used to compute a p-value for the null hypothesis that there is no difference between the survival curves in the two groups.

Similarly, W^2 is approximately a χ^2 -distributed statistic with 1 degree of freedom.

In our example:

```
fit.surv2<-survdif(Surv(time,status)~sex,data=BrainCancer)
fit.surv2
```

Call:

```
survdif(formula = Surv(time, status) ~ sex, data =
BrainCancer)
```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
sex=Female	45	15	18.5	0.676	1.44
sex=Male	43	20	16.5	0.761	1.44

Chisq= 1.4 on 1 degrees of freedom, p= 0.2

21.2 Regression models with survival response

Assume now we wish to predict the survival time T with features X .

Distribution and density (if continuous)

$$F(t) = P(T \leq t) = 1 - S(t), \quad f(t) = F'(t)$$

$P(T > t) = S(t)$ density of the time to event

The hazard function (instantaneous risk), also known as the force of mortality is formally defined as:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t}$$

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

It is the death rate in the instant after time t , given survival past that time.

Relationship:

$$h(t) = \frac{f(t)}{S(t)}$$

$$f(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t)}{\Delta t}$$

Cumulative hazard

$$H(t) = \int_0^t h(u) du$$

Key identity:

$$S(t) = \exp\{-H(t)\}$$

21.3 Likelihood for right-censored data

For independent subjects, the contribution of subject i is:

If event observed ($\delta_i = 1$): probability density at the event time $f(Y_i)$
 censored ($\delta_i = 0$): probability of surviving past censoring time $S(Y_i)$

So the full likelihood is

$$L(\theta) = \prod_{i=1}^n [f(y_i; \theta)]^{\delta_i} [S(y_i; \theta)]^{1-\delta_i} = \prod_{i=1}^n h(y_i)^{\delta_i} S(y_i)$$

$\times \frac{S(y_i; \theta)^{\delta_i}}{S(y_i; \theta)^{\delta_i}}$

and with $S(t) = \exp\{-H(t)\}$,

$$\log L(\theta) = \sum_{i=1}^n \delta_i \log h(Y_i; \theta) - \sum_{i=1}^n H(Y_i; \theta).$$

Parametric models: Exponential, Weibull, log-normal, etc., fit by maximizing the full likelihood above.

The Cox proportional hazards model assumes that

$$h(t | x) = h_0(t) \exp(\beta^\top x)$$

$\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots)$
regression part.

where $h_0(t) \geq 0$ is an unspecified function, known as the baseline hazard.

estimates covariate effects via partial likelihood (baseline hazard left unspecified).

For a one-unit increase in covariate x_j ,

$$\text{HR} = \exp(\beta_j).$$

$\text{HR} > 1$: higher hazard (shorter survival on average) $\text{HR} < 1$: lower hazard (longer survival on average)

For two covariate vectors x_a, x_b ,

$$\frac{h(t | x_a)}{h(t | x_b)} = \exp\{\beta^\top (x_a - x_b)\},$$

which is constant in time $\Rightarrow$ "proportional hazards".

Why it's useful

Separates time pattern of risk ($h_0(t)$) from covariate effects (β). Does not require specifying $h_0(t)$ (in Cox PH), yet estimates hazard ratios. Handles right-censoring naturally.

Estimation (Cox model idea) Cox estimates β via partial likelihood, using the ordering of event times. At each event time t_i , the probability that person i fails given someone in the risk set fails is

$$\frac{\exp(\beta^\top x_i)}{\sum_{j \in R(t_i)} \exp(\beta^\top x_j)}.$$

Multiplying over all events gives the partial likelihood:

$$L_p(\beta) = \prod_{d=1}^D \frac{\exp(\beta^\top x_{i(d)})}{\sum_{j \in R(t_{(d)})} \exp(\beta^\top x_j)}.$$

$e^{\beta^\top}$

To estimate β , we simply maximize the partial likelihood.

Baseline survival and prediction Once $\hat{\beta}$ is estimated, you can estimate baseline cumulative hazard $\hat{H}_0(t)$ (e.g., Breslow estimator) and then

$$\hat{S}(t | x) = \exp\{-\hat{H}_0(t) \exp(\hat{\beta}^\top x)\}.$$

Main assumption in most standard analyses: non-informative (independent) censoring, meaning censoring is unrelated to the future event time given covariates.

For our example:

```
fit <- coxph(Surv(time, status) ~ sex, data = BrainCancer)
summary(fit)
```

Call:

```
coxph(formula = Surv(time, status) ~ sex, data = BrainCancer)
```

n= 88, number of events= 35

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexMale	0.4077	1.5033	0.3420	1.192	0.233

	exp(coef)	exp(-coef)	lower .95	upper .95
sexMale	1.503	0.6652	0.769	2.939

Concordance= 0.565 (se = 0.045)

Likelihood ratio test= 1.44 on 1 df, p=0.2

Wald test = 1.42 on 1 df, p=0.2

Score (logrank) test = 1.44 on 1 df, p=0.2

Regardless of which test we use, we see that there is no clear evidence for a difference in survival between females and males.

Now we fit a model with all possible predictors

```
fit.all<-coxph(Surv(time,status)~sex+diagnosis+loc+ki+gtv+stereo)
fit.all
```

Call:

```
coxph(formula = Surv(time, status) ~ sex + diagnosis + loc +
      ki + gtv + stereo, data = BrainCancer)
```

	coef	exp(coef)	se(coef)	z	p
sexMale	0.18375	1.20171	0.36036	0.510	0.61012
diagnosisLG glioma	0.91502	2.49683	0.63816	1.434	0.15161
diagnosisHG glioma	2.15457	8.62414	0.45052	4.782	1.73e-06
diagnosisOther	0.88570	2.42467	0.65787	1.346	0.17821
locSupratentorial	0.44119	1.55456	0.70367	0.627	0.53066
ki	-0.05496	0.94653	0.01831	-3.001	0.00269
gtv	0.03429	1.03489	0.02233	1.536	0.12466
stereoSRT	0.17778	1.19456	0.60158	0.296	0.76760

Likelihood ratio test=41.37 on 8 df, p=1.776e-06

n= 87, number of events= 35

(1 observation deleted due to missingness)

diagnosis

The diagnosis variable has been coded so that the baseline corresponds to meningioma. The results indicate that the risk associated with HG glioma is more than eight times the risk associated with meningioma. After adjusting for the other predictors, patients with HG glioma have much worse survival compared to those with meningioma. In addition, larger values of the Karnofsky index, ki, are associated with lower risk, i.e. longer survival.

$$D = \frac{e^{-l}}{1 + e^{-l}}$$

```

modeldata<-data.frame(
  diagnosis=levels(BrainCancer$diagnosis),
  sex=rep("Female",4),
  loc=rep("Supratentorial",4),
  ki=rep(mean(BrainCancer$ki),4),
  gtv=rep(mean(BrainCancer$gtv),4),
  stereo=rep("SRT",4),
  data=BrainCancer)

survplots<-survfit(fit.all,newdata=modeldata)
plot(survplots,xlab="Months",ylab="Survial Probability",col=2:5,
legend("bottomleft",levels(BrainCancer$diagnosis),col=2:5, lty=:

```

